



2807 West County Road 75  
Monticello, MN 55362

May 17, 2023

L-MT-23-023  
10 CFR 50.73

ATTN: Document Control Desk  
U.S. Nuclear Regulatory Commission  
Washington, DC 20555-0001

Monticello Nuclear Generating Plant  
Docket No. 50-263  
Renewed Facility Operating License No. DPR-22

LER 2023-001-00 "Main Steam Isolation Valve Leakage Exceeds Technical Specification Requirements Due to Stem Scoring"

Pursuant to 10 CFR 50.73(a)(2)(i)(B), Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy (hereafter "NSPM"), hereby submits Monticello Nuclear Generating Plant (MNGP) Licensee Event Report (LER) 2023-001-00.

If you have any questions about this submittal, please contact Rick Loeffler, Senior Nuclear Regulatory Engineer, at [rick.a.loeffler@xcelenergy.com](mailto:rick.a.loeffler@xcelenergy.com) or at 612-342-8981.

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

A handwritten signature in black ink, appearing to read 'Christopher P. Domingos'.

Christopher P. Domingos  
Site Vice President, Monticello and Prairie Island Nuclear Generating Plants  
Northern States Power Company — Minnesota

Enclosure (1)

cc: Administrator, Region III, USNRC  
Project Manager, Monticello, USNRC  
Resident Inspector, Monticello, USNRC  
State of Minnesota

**ENCLOSURE 1**

**MONTICELLO NUCLEAR GENERATING PLANT  
LICENSEE EVENT REPORT 50-263/2023-001-00**

3 pages follow



## LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to [Infocollections.Resource@nrc.gov](mailto:Infocollections.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: [oir\\_submission@omb.eop.gov](mailto:oir_submission@omb.eop.gov). The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

|                                     |                                                                         |                  |         |
|-------------------------------------|-------------------------------------------------------------------------|------------------|---------|
| 1. Facility Name                    | <input checked="" type="checkbox"/> 050<br><input type="checkbox"/> 052 | 2. Docket Number | 3. Page |
| Monticello Nuclear Generating Plant |                                                                         | 263              | 1 OF 3  |

|                                                                                                     |
|-----------------------------------------------------------------------------------------------------|
| 4. Title                                                                                            |
| Main Steam Isolation Valve Leakage Exceeds Technical Specification Requirements Due to Stem Scoring |

| 5. Event Date |     |      | 6. LER Number |                   |              | 7. Report Date |     |      | 8. Other Facilities Involved |                              |
|---------------|-----|------|---------------|-------------------|--------------|----------------|-----|------|------------------------------|------------------------------|
| Month         | Day | Year | Year          | Sequential Number | Revision No. | Month          | Day | Year | Facility Name                | Docket Number                |
| 03            | 04  | 2023 | 2023          | - 001 -           | 00           | 05             | 17  | 2023 |                              | <input type="checkbox"/> 050 |
|               |     |      |               |                   |              |                |     |      |                              | <input type="checkbox"/> 052 |

|                   |                 |
|-------------------|-----------------|
| 9. Operating Mode | 10. Power Level |
| 1                 | 099             |

## 11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

|                                                    |                                                    |                                                       |                                             |                                                    |                                     |
|----------------------------------------------------|----------------------------------------------------|-------------------------------------------------------|---------------------------------------------|----------------------------------------------------|-------------------------------------|
| <input checked="" type="checkbox"/> 10 CFR Part 20 | <input type="checkbox"/> 20.2203(a)(2)(vi)         | <input checked="" type="checkbox"/> 10 CFR Part 50    | <input type="checkbox"/> 50.73(a)(2)(ii)(A) | <input type="checkbox"/> 50.73(a)(2)(viii)(A)      | <input type="checkbox"/> 73.1200(a) |
| <input type="checkbox"/> 20.2201(b)                | <input type="checkbox"/> 20.2203(a)(3)(i)          | <input type="checkbox"/> 50.36(c)(1)(i)(A)            | <input type="checkbox"/> 50.73(a)(2)(ii)(B) | <input type="checkbox"/> 50.73(a)(2)(viii)(B)      | <input type="checkbox"/> 73.1200(b) |
| <input type="checkbox"/> 20.2201(d)                | <input type="checkbox"/> 20.2203(a)(3)(ii)         | <input type="checkbox"/> 50.36(c)(1)(ii)(A)           | <input type="checkbox"/> 50.73(a)(2)(iii)   | <input type="checkbox"/> 50.73(a)(2)(ix)(A)        | <input type="checkbox"/> 73.1200(c) |
| <input type="checkbox"/> 20.2203(a)(1)             | <input type="checkbox"/> 20.2203(a)(4)             | <input type="checkbox"/> 50.36(c)(2)                  | <input type="checkbox"/> 50.73(a)(2)(iv)(A) | <input type="checkbox"/> 50.73(a)(2)(x)            | <input type="checkbox"/> 73.1200(d) |
| <input type="checkbox"/> 20.2203(a)(2)(i)          | <input checked="" type="checkbox"/> 10 CFR Part 21 | <input type="checkbox"/> 50.46(a)(3)(ii)              | <input type="checkbox"/> 50.73(a)(2)(v)(A)  | <input checked="" type="checkbox"/> 10 CFR Part 73 | <input type="checkbox"/> 73.1200(e) |
| <input type="checkbox"/> 20.2203(a)(2)(ii)         | <input type="checkbox"/> 21.2(c)                   | <input type="checkbox"/> 50.69(g)                     | <input type="checkbox"/> 50.73(a)(2)(v)(B)  | <input type="checkbox"/> 73.77(a)(1)               | <input type="checkbox"/> 73.1200(f) |
| <input type="checkbox"/> 20.2203(a)(2)(iii)        |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(A)            | <input type="checkbox"/> 50.73(a)(2)(v)(C)  | <input type="checkbox"/> 73.77(a)(2)(i)            | <input type="checkbox"/> 73.1200(g) |
| <input type="checkbox"/> 20.2203(a)(2)(iv)         |                                                    | <input checked="" type="checkbox"/> 50.73(a)(2)(i)(B) | <input type="checkbox"/> 50.73(a)(2)(v)(D)  | <input type="checkbox"/> 73.77(a)(2)(ii)           | <input type="checkbox"/> 73.1200(h) |
| <input type="checkbox"/> 20.2203(a)(2)(v)          |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(C)            | <input type="checkbox"/> 50.73(a)(2)(vii)   |                                                    |                                     |

☐ OTHER (Specify here, in abstract, or NRC 366A).

## 12. Licensee Contact for this LER

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Licensee Contact                                  | Phone Number (Include area code) |
| Rick Loeffler, Senior Nuclear Regulatory Engineer | 612-342-8981                     |

## 13. Complete One Line for each Component Failure Described in this Report

| Cause | System | Component | Manufacturer | Reportable to IRIS | Cause | System | Component | Manufacturer | Reportable to IRIS |
|-------|--------|-----------|--------------|--------------------|-------|--------|-----------|--------------|--------------------|
| B     | SB     | ISV       | A391         | Y                  |       |        |           |              |                    |

## 14. Supplemental Report Expected

|                                        |                                                                              |                              |       |     |      |
|----------------------------------------|------------------------------------------------------------------------------|------------------------------|-------|-----|------|
| <input checked="" type="checkbox"/> No | <input type="checkbox"/> Yes (If yes, complete 15. Expected Submission Date) | 15. Expected Submission Date | Month | Day | Year |
|                                        |                                                                              |                              |       |     |      |

## 16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

A packing leak on valve AO-2-86C, "13 Outboard Main Steam Isolation Valve" (MSIV), exceeded the Technical Specification (TS) individual MSIV leakage limit of Surveillance Requirement (SR) 3.6.1.3.12 on March 4, 2023, based on subsequent evaluation. The condition was discovered on March 21, 2023, when it was concluded that a reasonable expectation of operability was not met, and the MSIV was declared inoperable in accordance with TS 3.6.1.3 Condition E. During a maintenance outage on March 26, 2023, the Local Leak Rate Test as-found leakage was 987.1 standard cubic feet per hour (scfh) which exceeded the TS SR limit for individual MSIV leakage. This condition is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by the TSs. The cause of the packing leakage is attributed to scoring on the valve stem. During the planned maintenance outage, the MSIV was repacked to restore compliance with TS, and the valve packing gland was bored out and the stem was replaced during the 2023 Refuel Outage.



## LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
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|                                     |                                                                                              |                         |             |                         |                |
|-------------------------------------|----------------------------------------------------------------------------------------------|-------------------------|-------------|-------------------------|----------------|
| <b>1. FACILITY NAME</b>             | <input checked="checked" type="checkbox"/> <b>050</b><br><input type="checkbox"/> <b>052</b> | <b>2. DOCKET NUMBER</b> | <b>YEAR</b> | <b>3. LER NUMBER</b>    | <b>REV NO.</b> |
| Monticello Nuclear Generating Plant |                                                                                              | 263                     | 2023        | - SEQUENTIAL NUMBER 001 | - 00           |

### NARRATIVE

#### EVENT DESCRIPTION

While in Mode 1 at approximately 99 percent power, a packing leak being monitored on valve AO-2-86C, "13 Outboard Main Steam Isolation Valve" (MSIV), (i.e., the outboard MSIV on the "C" main steam line, hereafter referred to as the C outboard MSIV) exceeded the Technical Specification (TS) individual MSIV leakage limit of Surveillance Requirement (SR) 3.6.1.3.12 on March 4, 2023, based on subsequent evaluation. On March 21, 2023, engineering concluded that a reasonable expectation of operability for the C outboard MSIV [SB][ISV] was not met and operators declared the valve inoperable in accordance with TS 3.6.1.3 Condition E. During a planned maintenance outage on March 26, 2023, the 10 CFR 50, Appendix J Local Leak Rate Test (LLRT) as-found leakage for the C outboard MSIV was 987.1 standard cubic feet per hour (scfh) which exceeded the 100 scfh limit of SR 3.6.1.3.12 for individual MSIV leakage (when tested at greater than or equal to 44.1 psig). This condition is being reported within 60 days from the date of discovery, March 21, 2023, in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by the TSs as discussed in the event analysis.

#### Chronology

In May 2021, the as-left LLRT leakage from the 2021 Refuel Outage (RFO) for the C outboard MSIV was 33.9 scfh. In June 2021 a packing leak on this valve was identified and an Adverse Condition Monitoring Plan established to monitor the leak. On March 4, 2023, packing material was observed on camera being ejected from the packing gland on this valve. On March 10, 2023, a four-degree Fahrenheit step change increase in main steam chase temperature occurred. On March 14, 2023, at 1645 hours, due to an increase in leakage, operators entered the procedure for steam leaks outside primary containment, and on March 16, 2023, a prompt operability determination was requested. On March 18, 2023, the C inboard MSIV was closed to isolate the leak. On March 21, 2023, engineering concluded that a reasonable expectation of operability for the C outboard MSIV was not met due to the estimated leakage, and operations declared the valve inoperable in accordance with the TS 3.6.1.3 Condition E. It was concluded that March 4, 2023, was likely the date the TS individual MSIV leakage limit had been exceeded. On March 26, 2023, during the maintenance outage the as-found LLRT for the C outboard MSIV was 987.1 scfh which exceeded the 100 scfh limit of SR 3.6.1.3.12 for individual MSIV leakage (when tested at greater than or equal to 44.1 psig). Following repacking of the C outboard MSIV during this outage an as-left LLRT leakage rate of 67.7 scfh was measured. The C main steam line was returned to service with startup from the maintenance outage until the 2023 RFO began. During the 2023 RFO the C outboard MSIV packing gland was bored out to provide additional clearance and the valve stem was replaced, and the valve tested satisfactorily.

#### EVENT ANALYSIS

The event was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(i)(B), "Any operation or condition which was prohibited by the plant's Technical Specifications." Specifically, this occurrence is reportable as a condition prohibited by Specification 3.6.1.3, "Primary Containment Isolation Valves," because the MSIV had exceeded the individual leakage limit and had been inoperable for greater than the 8 hour Completion Time of Condition E and for greater than the Completion Times specified in Limiting Condition for Operation (LCO) 3.0.3, i.e., to be in Mode 2 within 7 hours, Mode 3 within 13 hours, and Mode 4 in 37 hours, respectively, when Condition E was not met.

This event is not classified as a safety system functional failure since the C inboard MSIV was fully operational.



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## 1. FACILITY NAME

Monticello Nuclear Generating Plant



050



052

## 2. DOCKET NUMBER

263

## 3. LER NUMBER

YEAR

2023

SEQUENTIAL

NUMBER

001

REV

NO.

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## NARRATIVE

## SAFETY SIGNIFICANCE

Each of the four main steam lines have an inboard and an outboard MSIV. The MSIVs safety function is to close in the event of a main steam line break and provide containment isolation. Two valves are provided in series to ensure the safety function is met even if one valve were to fail to close.

There were minimal safety consequences associated with this condition. During this event there was no equipment out of service or otherwise inoperable that contributed to the leakage. The MSIV leakage rate is assumed to be the actual pathway leakage rate for the flow path. The C inboard MSIV as-left leakage rate was 12.70 scfh during the 2021 RFO (the last outage prior to the planned maintenance outage) and the as-found leakage rate at the end of the two year operating cycle was 25.0 scfh for the 2023 RFO. The C inboard MSIV would have closed on an isolation signal ensuring the C main steam line flow path was isolated and within analyzed leakage limits.

The main steam pathway total minimum pathway leakage following the 2021 RFO was 67.811 scfh which is less than the SR 3.6.4.3.13 limit of less than or equal to 200 scfh (when tested at greater than or equal to 44.1 psig). Therefore, the primary containment isolation capability of the main steam lines remained operable which ensured the required isolation safety function was maintained.

## CAUSE

The cause of the packing leakage for the C outboard MSIV is attributed to scoring on the valve stem caused by contact with the packing gland follower.

## CORRECTIVE ACTION

During the planned maintenance outage, the C outboard MSIV (Anchor Darling 18-inch venturi, Model No. W9324183) was repacked and tested restoring compliance with TS leakage limits. During the 2023 RFO the valve stem was replaced, the packing gland was bored out to provide additional clearance, and the valve tested satisfactorily to meet TS leakage limits.

## PREVIOUS SIMILAR EVENTS

Monticello Nuclear Generating Plant LER 2017-003-00, "Main Steam Isolation Valve Leakage Exceeds Technical Specification Requirements," dated June 14, 2017, discusses where this same valve exceeded the TS individual MSIV leakage rate limit.

## ADDITIONAL INFORMATION

The Institute of Electrical and Electronics Engineer codes for equipment are denoted by [XX].